

Last Year Exam

Topics in the text book that are less relevant for the exam

- Chapter 3 – Deductive reasoning agents
- Chapter 6 – Understanding each other
- Chapter 13 – Forming coalitions
- Section 14.3 – Combinatorial auctions
- Section 15.4 – Bargaining for resource allocation
- Chapter 16 – Arguing
- Section 17.1.6 – Multiagent epistemic logics
- Section 17.2 – Logic of mental states
- Section 17.3 – Logic for cooperation

Topics in the Lecture Notes that are less relevant for the exam

- The Agora system

Grading

1. In order to pass exam (grade E) you have to get at least 36 points from questions from Part I (this includes your bonus points).
2. In order to get grade D you have to pass exam (see item 1) and get 49 points from all questions (this includes your bonus points).
3. In order to get grade C you have to pass exam (see item 1) and get 63 points from all questions (this includes your bonus points).
4. In order to get grade B you have to pass exam (see item 1) and get 74 points from all questions (this includes your bonus points).
5. In order to get grade A you have to pass exam (see item 1) and get 85 points from all questions (this includes your bonus points).

What is an agent?

1. If traffic lights (together with their control systems) are to be considered as intelligent agents, which of agent's properties should they employ and which of agent's properties they can't employ (if any)? Illustrate your answer by examples.

(6p)

Agent theory

2. What are relations between Beliefs, Desires, Goals and Intentions in the BDI-architecture?

(4p)

Agent Architectures

3. What is a purpose of introducing the “see” function into an abstract agent architecture? Give an example

(4p)

4. What are basic types of layered agent architectures for hybrid agents? Briefly explain their advantages and disadvantages

(4p)

Negotiation

5. Pareto efficiency is an important property in negotiation. Briefly explain it. What are relations between Pareto efficient solutions and social welfare maximizing solutions?

(4p)

6. If we compare the binary protocol and Borda protocol. Which of them is more computationally efficient? Which of them is not independent of irrelevant alternatives? Explain.

(4p)

7. We considered Clark tax algorithm. What is its objective? How it achieves this objective? (It is not necessary to present particular formulae in your answer)

(4p)

Communication

8. Having semantics of the request(s,h,a) speech act as it was defined in lecture notes, which of the following statements is/are precondition(s) and which of them is/are post-condition(s) of this speech act?

- h believes (s believes (s wants a))
- s believes (h believes (h can do a))
- s believes (s wants a)
- s believes (h can do a)

(3p)

9. Give basic ideas of the agent communication languages KQML and FIPA. What do they have in common and what are basic differences between them?

(4p)

Coordination

10. Explain main organizational structures that you know. Give their advantages and disadvantages.

(4p)

11. Give examples of social norms and laws in agent coordination. What is difference between them?

(3p)

Mobile agents

12. Give examples of situations (at least 2 situations) where mobile agents are appropriate (when they have benefits over communicating non-mobile agents)? Justify your answer.

(4p)

Part II (25 p.)

13. What are axioms of the logic of knowledge? Which axiom(s) from the logic of knowledge is/are not valid in the logic of belief and why?

(5p)

Part II (25 p.)

14. This is a payoff matrix for the Prisoners dilemma:

		Defect	Coop
j	Defect	2, 2	5, 0
	Coop	0, 5	3, 3

Which cells in this matrix correspond to: social welfare, Pareto optimal solution and Nash equilibrium? Explain your answer.

(5p)

Part II (25 p.)

15. What is/are the main difference(s) between meta-level information exchange coordination and multi-agent planning?
(5p)

16. What is/are main difference(s) between MAS architectures that use middle agents and marketbased MAS architectures?
(5p)

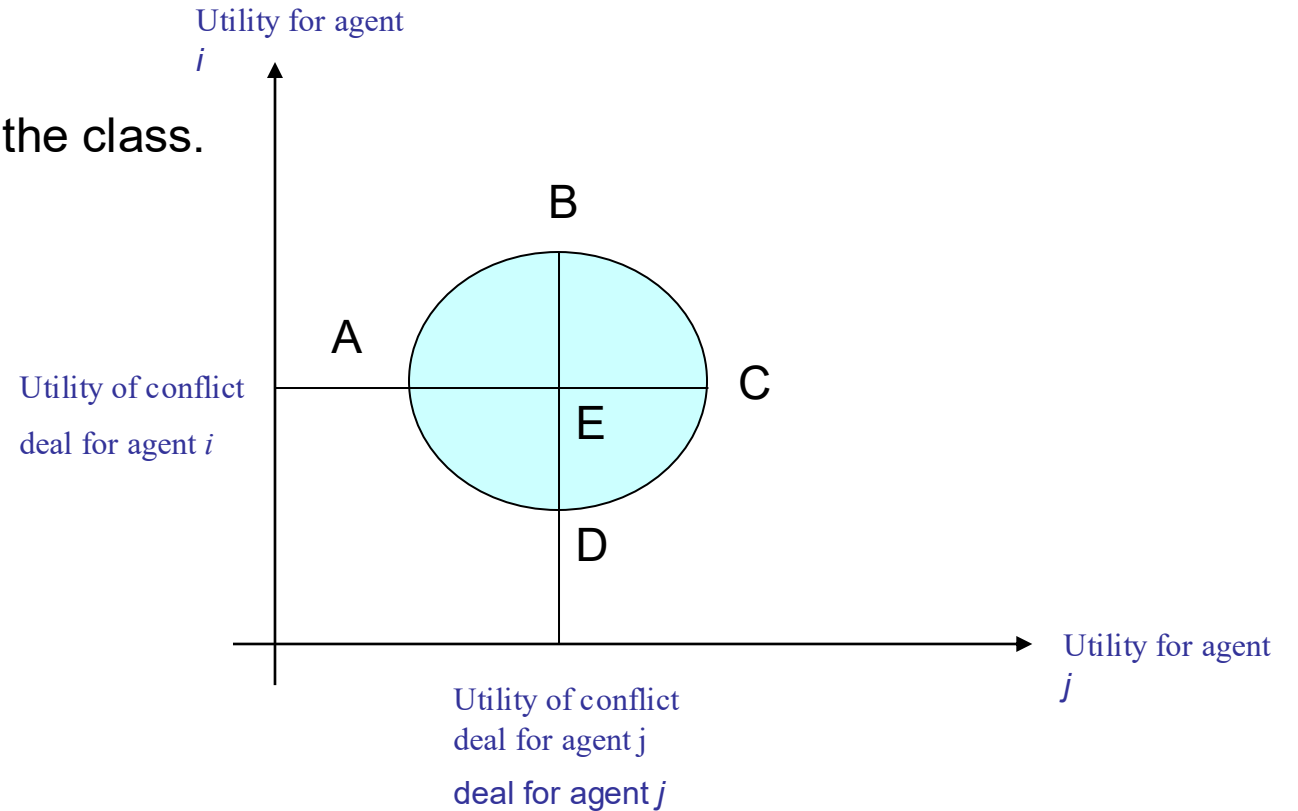
Part II (25 p.)

17. Show an implementation (in any (pseudo-)code) of action selection function for agents in the subsumption architecture?

(5p)

Part III (24 p.)

19. We considered the following figure in the class.



Show on the figure points/areas for:

Conflict deal, negotiation set, possible deals, Pareto optimal deals and individually rational deals.

Explain your answer.

(7p)

Part III (24 p.)

20. Is it possible that the same propositional content of the ACL message can be used to present different speech acts? Give an example if the answer is yes and justify if the answer is no.

(7p)

Part III (24 p.)

20. If we assume that the statement “agent doesn’t know whether p ” is equal to “ $\Diamond p \ \& \ \Diamond \neg p$ ” in modal logic. Express in the logic of knowledge the following statement:

“I don’t know whether I know the answer and I don’t know whether my neighbor knows the answer”

Explain your answer

(10p)